MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 4

(1) Determine if the following statements are true or false. Give a line or two in justification.

(a) The fields $\mathbb{F}_5[x]/(x^4+x+1)$ and $\mathbb{F}_5[x]/(x^4+x^2+x+1)$ are both isomorphic to each other. (*You can assume the given polynomials are irreducible in* $\mathbb{F}_5[x]$)

Solution. True. They are both fields of order $5^4 = 625$ and so are isomorphic by Galois' theorem.

(b) If k is a finite field of order 32, then $x^{64}-x$ has 32 solutions in k.

Solution. False. By part (c) of the Proposition from p. 2 of Lecture 10, there are exactly $2^{\gcd(5,6)} = 2^1 = 2$ solutions.

(c) There are exactly 36 irreducible polynomials of degree 4 in $\mathbb{F}_3[x]$.

Solution. True. There are 18 such polynomials with leading coefficient 1: By Problem 6 from Homework 3, this is the number of such polynomials of degree 4 dividing $\frac{x^{3^4}-x}{x^{3^2}-x}$, which as degree $81-9=72$. But we can change the leading coefficient to -1 as well, giving us 18 more.

(d) There is an irreducible cubic polynomial in $\mathbb{F}_5[x]$ with exactly one zero in a field of order 125.

Solution. False. By the Remark on p.5 of Lecture 9, any irreducible polynomial of degree 3 in $\mathbb{F}_5[x]$ has 3 distinct zeros in any field of order $125 = 5^3$.

(e) Every field of order 81 contains a field of order 9.

Solution. True. By part (c) of the Proposition from p. 2 of Lecture 10, if $|k| = 81 = 3^4$, then the fixed points of Fr^2 form a field k_m with order $|k_m| = 3^{\gcd(2,4)} = 3^2 = 9$.

(2) Suppose that $f(x) \in \mathbb{F}_p[x]$ is an irreducible polynomial of degree r and let k be a finite field of characteristic p. Show that the following are equivalent:

(a) $|k| = p^d$ where $r \mid d$;

(b) $f(x)$ admits a zero in k.

Hint: This is essentially a reformulation of problem 4(b) and (a)⇒(b) in problem 5 of Homework 3.

Solution. (a)⇒(b): This was actually shown in the Proposition from p. 5 of Lecture 9. The main point was that $f(x) \mid x^{p^d} - x$, which (a)⇒(b) in problem 5 of Homework 3.

(b)⇒(a): If $f(x)$ admits a zero $\alpha \in k$, then we obtain the usual factoring triangle

$$\begin{tikzcd} \mathbb{F}_p[x] \arrow[d] \arrow[dr, "\mathrm{ev}_\alpha :\ f(\alpha)=0"] & \\ \mathbb{F}_p[x]/(f(x)) \arrow[r, "\overline{\mathrm{ev}}_\alpha"'] & k \end{tikzcd}$$

The bottom arrow, since it originates from a field (uses the irreducibility of $f(x)$), is injective. This means that $\mathbb{F}_p[x]/(f(x))$ is isomorphic to its image, which is a subfield of k. Now, $|\mathbb{F}_p[x]/(f(x))| = p^r$ (why?) and $|k| = p^d$. By problem 4(b) of Homework 3, this means that $r \mid d$.

The next problem completes the proof of (b)⇒(a) of problem 5 of Homework 3.

(3) Let $f(x) \in \mathbb{F}_p[x]$ once again be an irreducible polynomial of degree r, and suppose that $f(x) \mid x^{p^d} - x$. Set $k = \mathbb{F}_p[x]/(f(x))$ and $\alpha = x + (f(x)) \in k$.

(a) Show that r is the smallest positive integer m such that $\mathrm{Fr}^m(\alpha) = \alpha$
Hint: Use the previous problem.

Solution. By part (c) of the Proposition from p. 2 of Lecture 10, the solutions β to $\mathrm{Fr}^m(\beta) = \beta$ form a subfield k_m of $\mathbb{F}_p[x]/(f(x))$ of size $p^{\gcd(r,m)}$. So, saying that $\mathrm{Fr}^m(\alpha) = \alpha$ is the same as saying that α belongs to k_m. However, by the previous problem (applied with $k = k_m$), this means that

$$r \mid \gcd(r, m).$$

The only way for this to be possible for $m \leq r$ is if $r = m$.

(b) Show that $\mathrm{Fr}^{\gcd(r,d)}(\alpha) = \alpha$.

Solution. Our hypothesis is that $f(x) \mid x^{p^d} - x$, meaning that $x^{p^d} - x = f(x)g(x)$ with $g(x) \in \mathbb{F}_p[x]$. Therefore,

$$\alpha^{p^d} - \alpha = f(\alpha)g(\alpha) = 0 \cdot g(\alpha) = 0 \ \Rightarrow\ \mathrm{Fr}^d(\alpha) = \alpha \Rightarrow \alpha \in k_d.$$

By the proof of part (c) of the Proposition from p. 5 of Lecture 10, we know

$$k_d = k_{\gcd(r,d)} = \{\beta :\ \mathrm{Fr}^{\gcd(r,d)}(\beta) = \beta\}.$$

Since $\alpha \in k_d$, this finishes the proof.

(c) Conclude that $r \mid d$.

Solution. Putting (a) and (b) together tells us that $\gcd(r, d) = r$, which is another way of saying $r \mid d$.

The previous problem completes our proof of Galois' Theorem:

Theorem 1. *For every prime p and every $d \geq 1$, there is a* unique *field of order p^d up to isomorphism.*

Notation 1. We will denote the unique field of order p^d by $\mathbb{F}_{p^d}$.

Caution 1. When $d \geq 2$, this is *very* different from $\mathbb{Z}/p^d\mathbb{Z}$. Do *not* confuse the two.

(4) Suppose that we have $\alpha \in \mathbb{F}_{p^d}$ and that $m_\alpha(x) \in \mathbb{F}_p[x]$ is its minimal polynomial over $\mathbb{F}_p$. Set $r = \deg m_\alpha(x)$.

(a) Show that $\mathrm{Fr}(\alpha) = \alpha^p$ also has minimal polynomial $m_\alpha(x)$.

Solution. The main point is that if we have an expression $\sum_i a_i\alpha^i = 0$ with $a_i \in \mathbb{F}_p$, then

$$0 = \mathrm{Fr}(\sum_i a_i\alpha^i) = \sum_i \mathrm{Fr}(a_i)\mathrm{Fr}(\alpha)^i = \sum_i a_i(\alpha^p)^i.$$

Here, we are using the fact that $\mathrm{Fr}(a) = a^p = a$ for all $a \in \mathbb{F}_p$.
What the above equation is telling us is that if we have a polynomial $g(x) \in \mathbb{F}_p[x]$ and $g(\alpha) = 0$, then $g(\mathrm{Fr}(\alpha)) = g(\alpha^p) = 0$.
Applying this to $m_\alpha(x)$ tells us that $m_\alpha(\mathrm{Fr}(\alpha)) = 0$. This means that $m_\alpha(x) \in \ker \mathrm{ev}_{\mathrm{Fr}(\alpha)}$, which in fact implies that $m_{\mathrm{Fr}(\alpha)}(x) = m_\alpha(x)$.

(b) In fact, show that the set

$$\{\mathrm{Fr}^m(\alpha) :\ 0 \leq m \leq r-1\} \subset \mathbb{F}_{p^d}$$

is exactly the set of r distinct zeros of $m_\alpha(x)$ in $\mathbb{F}_{p^d}$.

Solution. Iterating part (a) tells us that $\mathrm{Fr}^m(\alpha)$ is a zero for $m_\alpha(x)$. Now, suppose that we have $0 \le j \le i < r = \deg m_\alpha(x)$. Then $\mathrm{Fr}^i(\alpha) = \mathrm{Fr}^j(\alpha)$ is true if and only if

$$\mathrm{Fr}^{i-j}(\alpha) = \alpha.$$

This is using the fact that Fr is an automorphism so Fr^{-1} makes sense! Now, by problem 3(a), this is only possible if $i = j$ (why?). Therefore, we see that the elements $\{\alpha, \mathrm{Fr}(\alpha), \ldots, \mathrm{Fr}^{r-1}(\alpha)\}$ are all distinct. But they are all also zeros for $m_\alpha(x)$, which as degree r, and so must be *the* r distinct zeros of $m_\alpha(x)$ in $\mathbb{F}_{p^d}$.

(5) Show that the following are equivalent for $\alpha \in \mathbb{F}_{p^d}$ and an integer $r \ge 1$:
   (a) r is the smallest positive integer m such that $\mathrm{Fr}^m(\alpha) = \alpha$;
   (b) The smallest subfield of $\mathbb{F}_{p^d}$ containing α has order p^r;
   (c) $m_\alpha(x)$ has degree r.

Solution. Before we begin, note that, for all integers m, we have the subfield $k_m \le \mathbb{F}_{p^d}$ of elements β satisfying $\mathrm{Fr}^m(\beta) = \beta$. Moreover, the following properties are true:
- This subfield has size $p^{\gcd(d,m)}$;
- We also have $k_n \le k_m$ when $n \mid m$ (since Fr^m is an iterate of Fr^n in this case);
- The fields of the form k_m are *all* the subfields of $\mathbb{F}_{p^d}$ (why?).

(a)⇔(b): (a) is now saying that r is the smallest integer such that $\alpha \in k_r$. This is now clearly equivalent to (b): The only observation is that $r \mid d$, because otherwise we would be able to replace r with the smaller integer $\gcd(r, d)$.

(c)⇒(a): this is just 3(a) applied with $f(x) = m_\alpha(x)$.

(a)⇒(c): Looking at the proof of 4(b), we see that (a) is telling us that the elements $\{\mathrm{Fr}^i(\alpha) :\ 0 \le i \le r\}$ are the distinct zeros of $m_\alpha(x)$, and this implies that $m_\alpha(x)$ has degree r (why?).

(6) Suppose that $\alpha \in \mathbb{F}_{p^d}^\times$ is an element of order m. Show that

$$\deg m_\alpha(x) = \text{order of } p \text{ in } (\mathbb{Z}/m\mathbb{Z})^\times.$$

Note the reversal of the roles of p and m! This is an example of what is called a reciprocity law.

Solution. If k is a subfield of $\mathbb{F}_{p^d}$ containing α, then $k^\times$ contains an element of order m. But such an element exists if and only if $m \mid k^\times$. If $|k| = p^r$, then $m \mid p^r - 1$, which is equivalent to $p^r \equiv 1 \pmod{m}$. The smallest r for which this is possible is the order of p in $(\mathbb{Z}/m\mathbb{Z})^\times$! Now, problem 5 (namely the implication (b)⇒(c)) tells us that this is also the degree of $m_\alpha(x)$.

Definition 1. Let $\mathrm{Aut}(\mathbb{F}_{p^d}/\mathbb{F}_p)$ be the set of field automorphisms of $\mathbb{F}_{p^d}$. This is usually also denoted $\mathrm{Gal}(\mathbb{F}_{p^d}/\mathbb{F}_p)$, which is notation we will come back to later in the semester. This is a *group* under composition (why?).

Can you guess what Gal *stands for?*

(7) (a) Show that there is a unique group homomorphism $\mathbb{Z}/d\mathbb{Z} \to \mathrm{Aut}(\mathbb{F}_{p^d}/\mathbb{F}_p)$ that takes 1 to Fr;

   Solution. This follows from the homomorphism property of $\mathbb{Z}/d\mathbb{Z}$ and the fact that Fr^d is the identity.

   (b) Show that this gives a group action of $\mathbb{Z}/d\mathbb{Z}$ on $\mathbb{F}_{p^d}$ for which we obtain a bijection

$$\mathrm{Irr}_{|d}(\mathbb{F}_p) \xrightarrow{\simeq} \{\text{Orbits of the action of } \mathbb{Z}/d\mathbb{Z} \text{ on } \mathbb{F}_{p^d}\}$$

   Solution. This is simply a reinterpretation of 4(b).

We will see later in the term that the homomorphism in (a) is in fact an isomorphism.

(8) Combine the previous problem with Burnside's Lemma to show that, if N_r is the number of irreducible polynomials in $\mathbb{F}_p[x]$ of degree r with leading coefficient 1, then we have

$$\sum_{r|d} N_r = \frac{1}{d} \sum_{m=0}^{d-1} p^{\gcd(m,d)}.$$

Solution. Burnside's lemma says that the number of orbits for a group action $G \curvearrowright X$ is

$$\frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)|,$$

where $\mathrm{Fix}(g) = \{x \in X : \; g \cot x = x\}$.

Apply this with $X = \mathbb{F}_{p^d}$ and $G = \mathbb{Z}/d\mathbb{Z}$: 6(b) tells us that the number of orbits is exactly the size of $\mathrm{Irr}_{|d}(\mathbb{F}_p)$. This is just the quantity $\sum_{r|d} N_r$.

On the other hand, for $m \in \mathbb{Z}/d\mathbb{Z}$, $\mathrm{Fix}(m) = \{\alpha \in \mathbb{F}_{p^d} : \; \mathrm{Fr}^m(\alpha) = \alpha\} = k_m$, which has size $p^{\gcd(m,d)}$. Plugging this into the equation from Burnside's lemma now completes the proof.